c_1 1960 - 2018

$0, 0$ | $1, 2$

$b \to c_2$

$k=1$

$n=2$

2nd

1950 - 2018

$k=1$

$k=2$

$|S| \geq \frac{n}{2}$

c_3

$|T|=1$

$\boxed{c_1 \ c_2}$

$\boxed{c_1}$

$\boxed{c_2}$

$\frac{n}{2}$

$|T|=k$

$k=3$ $|T|=2$

$\frac{1}{2}$

$|T|=n$

JR

PAV

$\boxed{c_1 \ c_2 \ c_3}$

$\frac{2n}{3}$

$c'_1 \ c'_2$

$\frac{n}{3}$

PAV

$\frac{n}{3}$

1

$2k \times \binom{t}{2}$ $x_i x_j$ y_i

$\downarrow$

$kt(t-1)$ x_i x_j y_i

$k(k-t)(t-1)$ $y_1 - \boxed{y_k}$

$-(k-t)(t-1) + \binom{t-1}{2}$

t	$2k\binom{t}{2}$
$k-t$	$k(k-t)(t-1)$

$\binom{t-1}{2}$ 1 0

$(t-1)$ $2k-1$ 1

$k \rightarrow k(k-t)(t-1)H_k + kt(t-1)H_1$

$k\ 1 \rightarrow k(k-t)(t-1)H_{k-1} + (2k-1)(t-1)H_2 + \left((t-1) + (2k-1)\binom{t-1}{2}\right) \cdot H_1 + 0 \times \binom{t-1}{2}$

$(k-t)(t-1) + \binom{t-1}{2} \cancel{t} > \frac{(2k-1)(t-1)}{2}$

$\binom{t-1}{2} > (t-1)\left((k-\frac{1}{2}) - k + t\right)$

$\frac{t-2}{2} < \left(t - \frac{1}{2}\right)$

$x_i \; x_j$ $\quad 2 \; \frac{1}{3} \; \frac{4}{6} \; \frac{2}{3}$

$\frac{2}{3} \quad \frac{1}{3} \quad 0$

$18H_6 + 18$

$18H_5 + 10H_2 + 7$ $\quad \frac{18}{5} + 4 \quad \frac{13}{5}$

$18H_4 + 10H_2 + 4H_3 + 4$ $\quad 1 + \frac{4\times 5}{6} \quad \frac{26}{6}$

$x_t \quad y_{2t}$

$x_i \uparrow \quad y_i \downarrow \; 2t \times \binom{t}{2} \quad x_i \, x_j \quad y_i \quad \leftarrow b$

$\downarrow \; \frac{1}{k} \quad 2t \times \binom{t}{2} \; y_1 - y_{2t} \quad \leftarrow a$

$n \quad 2t^2(t-1) \, H_a + \binom{b}{2} a \, H_3 + \left(b(t-b)(2t-a) + \binom{t-b}{2} a\right) H_1 + b(t-b)a \, H_2$

$$\frac{\binom{t-b}{2} \times 2t}{4t \times \binom{t}{2}} \geq \frac{t-b}{2t} \iff \frac{(t-b)(t-b-1)}{t(t-1)} \geq \frac{t-b}{t}$$

$$\iff b = 0$$

EJR core

✓ ✗

1 c_1

3 d_i c_1 c_2

2 d_1 d_2 d_3

$|S| =$ d_i

$k \leftrightarrow m$

$|T| = l$

l

$|T|$ 7

$\frac{1}{\sqrt{l}}$

$\frac{2}{3}$

$\frac{10}{6} + 3$

$c_1 = 4$

2

$\frac{k}{2} H$

k $\frac{k}{2}$ x

$\frac{k}{2}$ y

$\frac{k}{2}$

$15H_6 + 15H_2$

$15H_3 + 3H_5$

$+ 9H_4 + 3H_3$

15

15

Approval CC

JR ✓

$|S| \geq \frac{n}{k}$

$c_i \in \cap A_i$

$c_i \rightarrow \frac{n}{k}$ افزایش

$\frac{n - \frac{n}{k}}{k}$ کاهش

PJR ×

$l = 2 \quad k = 3$

$c_1 \quad c_2$

c_3

c_4

$4 = |S| \geq \frac{2}{3} 6 = 4$

$|A_i| \geq 2$

$\cup A_i = \{c_1, c_2\}$

JR < PJR مثال برای

PJR → ⟹ Preserve EJR

EJR ×

PJR

PAV EJR

$|S| \geq \frac{ln}{k}$

$|\cap A_i| \geq l$

$\frac{n|T|}{k} \leq |S|$ S'

$\frac{1}{3}$

$\frac{a}{a+5b} = \frac{3}{5}$

$|A_i \cap T| > |A_i \cap W|$

$2a = 15b$

$a = 15$

$|T| \to \frac{1}{|T|} \times |S| \geq \boxed{\frac{n}{k}}$

$S \to a \quad c'_1 \quad c'_2 \quad c'_3$

$2 \quad b \quad c_1$

$2 \quad c_2$

$2 \quad c_3$

$2 \quad c_4$

$2 \quad c_5$

$c_1 \ldots c_k$ $\boxed{k - |T|}$

$\frac{3}{5}$

$\frac{n-|S|}{}$

$k=5$

$|T| = 3$

$5D$

2

5

$\left(\frac{n-|S|}{k}\right)|T| \leq \frac{1}{|T|} \times |S|$

$5, 3$

$\leq \frac{n}{k}$

$\left(\frac{n}{k} - \frac{n}{k^2}|T|\right)|T| \leq \frac{n}{k} \leq$

$|T| + 1 - \frac{|T|}{n}$

$1 - \frac{2|T|}{k}$

2

$\frac{2}{5}\left(1 - \frac{|T|}{k}\right)|T| \leq 1$

$\frac{k}{2}\left(1 - \frac{1}{k}\right)$

$\frac{k}{2}$

$k \binom{t}{2}$ x_i x_j y_i $\frac{t}{k}$

$\frac{k(k-t)(t-1)}{2}$ $y_1 \dots y_k$ $\frac{k+(t-1)}{2}$

$k(k-t)(t-1$

$c_1 - c_k$ $d_1 - d_k$ 0 $\frac{n}{k}$

c_1 $\vdots$ c_k $d_1 \dots d_{\frac{k}{2}}$ $\frac{n}{2}$ $\frac{n}{2}$

$a\binom{k}{\frac{k}{2}-1}$

$H_{\frac{k}{2}} - H$

$a\binom{k}{\frac{k}{2}-1} = bk$

$\frac{k}{2(k+1)}\binom{k+1}{\frac{k}{2}}$

$k=4$

a

b

c

d

aef

bef

cef

def

ab $\rightarrow$ ef

abef

ef.

PAV

$\frac{k}{2}$

6

5

≥ 6

12

6

20

ab $\frac{1}{6}$

$|T|=2$

$\rightarrow \frac{4}{2}$

$C_3 \cdots C_{12} \rightarrow \frac{1}{20}$

$C_1 \rightarrow \frac{1}{8}$

$C_2 \rightarrow \frac{1}{13}$

$a, b \rightarrow \frac{1}{4}$

$\frac{6}{12} + \frac{6}{11} + \frac{5}{11}$

$\boxed{\frac{1}{8} + \frac{1}{13} + \frac{1}{2}}$

$\frac{1}{4} + \frac{1}{13}$

$\frac{a+c}{a+b}$ $\frac{b-c}{a+b}$

$\frac{(b-c)}{(a+b)b} > \frac{(a+c)}{(a+b)c(a+1)}$

$b > \frac{a+1}{a} \frac{c^2}{c-1}$

$c=2$ 4

$k W_{k-a} > \frac{(a+c)}{\cancel{c}} \left(\frac{W_{a+1} + cW_{k-a}}{c} \right)$

$\frac{a}{c} X +$ $\frac{a}{c} X + cY$

$\downarrow$

PAV $\frac{1}{1}$ $\frac{1}{2}$ $\frac{1}{3}$, ...

$w_1 > w_2 > \dots$

هر وزن

PAV

خراب است:

$\frac{b-c}{\cancel{a+b}} w_b > \frac{a+c}{\cancel{(a+b)}c} w_{a+1}$

$b w_b > c w_b + \frac{a+c}{c} w_{a+1}$

$(b-c) c W_b > (a+c) W_{a+1}$

k

$(a-c+1)c > (a+c)$

$c=2$

$a=99$

$b=100$

$200 > 101$

$(k-(a+c)) c W_{k-a} > (a+c) W_{a+1}$

$k=9$ $a=4$ $\rightarrow$ $9-(c+4)$ 2b-1 t

$(5-c)c > 4+c$

$kcW_{k-a} > (a+c)(W_{a+1} + cW_{k-a})$

$\frac{(a+b)c}{b} > (a+c)\left(\frac{b}{a+1} + \frac{c}{b} \right)$

$b((a+1)^c - (a+c)) > (a+c)(a+1)c - a(a+1)$

$b(1-c) > (a+1)(c^2 + ac - a)$

$k\, W_{k-a} > (a+c)\left(\frac{W_{a+1}}{c} + W_{k-a}\right) \rightarrow$ انتخاب وزن بر حسب k

$k > (a+c)\left(\frac{W_{a+1}}{(W_{k-a})c} + 1\right)$

$a+c < k$

$\frac{1}{2^2}$

2^{k-1}

$k \leq (a+c)\left(\frac{\varepsilon^{1-k}}{c} + 1\right)$

$k >$

$\frac{c+1}{c}\, \varepsilon^{1-k} + c$

$k \leq (a+1)\left(\frac{w_{a+1}}{w_{k-a}} + 1\right)$

2 1

$(k-1)\, w_k < w_1$

$k \leq (a+c)\left(\frac{W_{a+1}}{w_{k-a}\, c} + 1\right)$

$\frac{k-1}{k}$ $\frac{1}{k}$

$(k-1)\, w_k < w_1$

$k \leq (k-1-a-c)\left(\frac{W_{k-a}}{W_{a+1}}\, c + 1\right) \rightarrow ?$

$w_k < \frac{1}{k-1}$

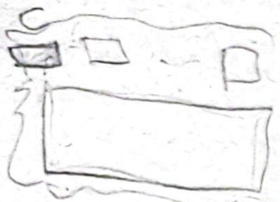
$k^2 \leq$

$\frac{k-a}{k}$ ε^k

$\varepsilon^k = \frac{1}{k}$

$b > \frac{a+1}{a}$ $\frac{c^2}{c-1}$

$c=4$ $a=\cancel{2}$ $b=17$ 998



$\frac{a+c}{a+b}$ $\frac{b-c}{a+b}$

$\frac{1}{13}$ $\frac{1}{k}$

$\frac{1}{6}$ $\frac{6}{19}$ $\frac{13}{19}$ $\frac{0.994}{100}$

0.006

$\frac{998}{994}$

$a+b$ $b-$ $\frac{2}{6}$

$994\left(\frac{996}{994}\right)^2 + \frac{8}{3}$

$\boxed{max\,min}$
$min\,max$

$\frac{b-c}{a+b}$

$994+6$

8 $994\left(\frac{2}{994} \times \frac{998}{994}\right.$

$a\,bc$ 1

$c \in T$ $\sum x_i^2$

$a\,\boxed{ef}$ 15 $994($

$c' \in W/T$ $\frac{k}{n}$

$\geq \frac{n}{k}$ T

b e d $\boxed{ef}$ k

$994\left(\frac{995}{994}\right)^2 + 3x$

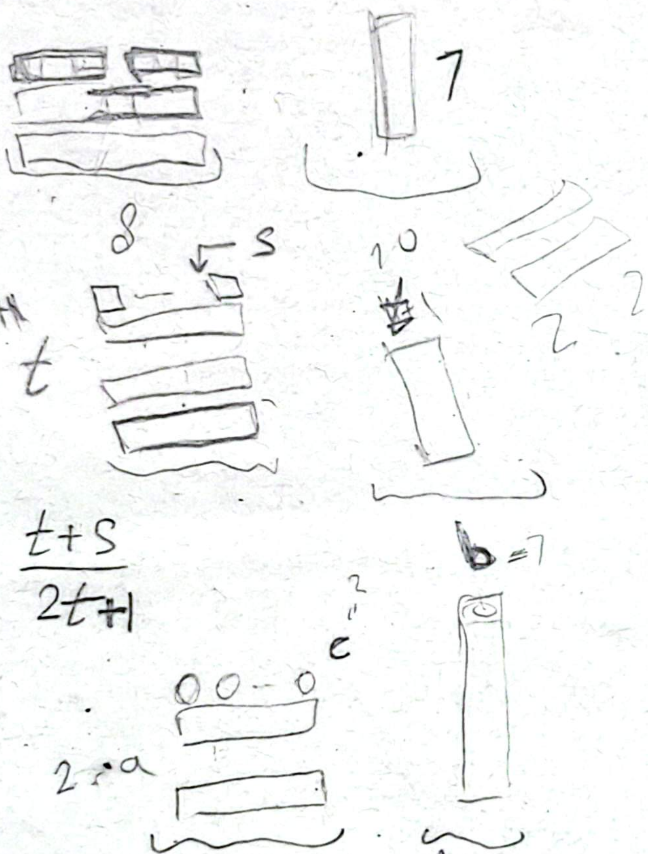
c c

c' c'

IDK

$k=10$ 3 $\frac{7}{10}$ $\frac{4}{10}$

$n=20$



$k=2t+1$

$\frac{t+s}{2t+1}$

$\frac{a+c}{b+c}$

$\frac{b-a}{b}$

$\frac{12}{27}=\frac{4}{9}$

16 13

13

c/a $\frac{2}{3}$ $\frac{5}{7}$

$(b-a)(a+1)c > b(a+c)$

$b(ac-a) > ac(a+1)$

$ba(c-1) > ac(a+1)$

$\frac{b(c-1)}{c} > (a+1)$

$b > \frac{c}{c-1}(a+1)$

$k=a+b$

$c=2 \rightarrow b > 2(a+1)$

$b > \frac{10}{9}\frac{(a+1)}{2}$

b

$\frac{b-a}{b} > \frac{a+c}{c^2}$

$\frac{(b-a)c}{b} > \frac{a+c}{a+1}$

$(b-a)c > b(a+c)$

$c^2(b-a) > b(a+c)$

$a+b+c$

$b(c^2-a) > \ c^2a$

$b(c^2-c-a) > 4a$

$b(2-a) >$

$a=1$

$b=5$

$\frac{2\times 3}{5}$

8 5 6

PAV

$|A_i \cap W| < |A_i \cap T|$

$c \in T/W$ $\quad$ $W \cup \{c\}$

c'

$\underbrace{\sum x_i^2}$

$c(a+c)$ $\quad$ $c(b$ $\quad$ $c=2$ $\quad$ 3

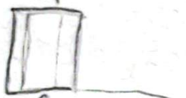
$\frac{a+c}{a+b}$ $\quad$ $\frac{b-c}{a+b}$ $\quad$ $\frac{b-c}{a+b}$ $\quad$ $\frac{b-1}{a+b}$

$2\sqrt{4}$

$n\frac{|T|}{k} \leq |S|$

8

160

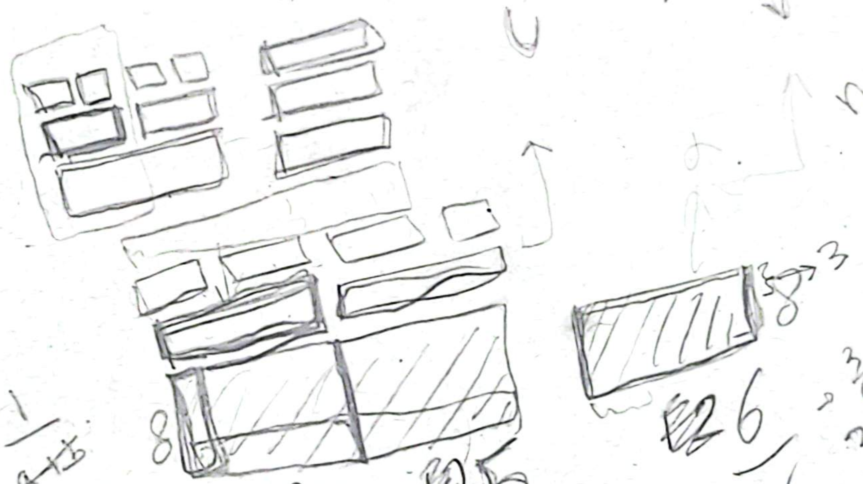


$\frac{1}{12}$

12

96

$\frac{8}{96}$



8

$\frac{5}{16}$ $\quad$ $\frac{5}{16}$ $\quad$ $\frac{6}{16}$

3 → 3

8

$\frac{3}{8}$ $\quad$ $\frac{3}{4}$

$c \in T$ $\quad$ 8 $\quad$ b

m/00

$\frac{b-c}{b(a+b)} < \frac{1}{a+b}$

c

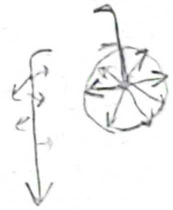
a

$\frac{a+c}{a+b}$

b

$\frac{b-c}{a+b}$

$\frac{3}{8}$ $\quad$ $\frac{8}{16}$ $\quad$ $\frac{1}{k}$



$\frac{m}{64}$

$\frac{1}{16}$

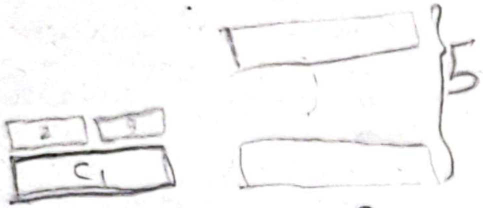
k=16 $\quad$ k=16

b=8

a=8

c=1

PAV



3 3 8

$\frac{a+c}{a+b}$

$\frac{8}{5}$ $\frac{3}{2}$ $3/6$

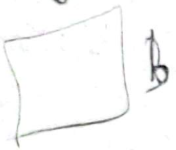
$$\boxed{b > \frac{a+1}{a} \quad \frac{c^2}{c-1}}$$

4

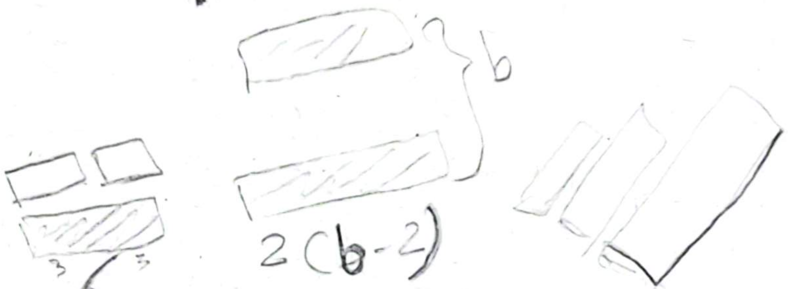
$c=2$

$a=1$

$b>8$



$c(a+c)$ $c(b-c)$



6 $2(b-2)$

$$\frac{c(b-c)}{b} > \frac{a+c}{a+1}$$

$$2\frac{(b-2)}{b} > \frac{3}{2}$$

$4b$

$a=8$

$b=8$

8 20 8 12

$$b(ac+c-a-c) > c^2(a+1)$$

$$ba(c-1) > c^2(a+1) \quad c=2$$

$c=1$ $a=2$ $b=2$

$\frac{1}{2}$ $\frac{3}{2}$

3 1

20 $\frac{1}{20}$

$x_1 \quad x_2 \quad x_3 \quad \binom{6}{2}$

$y_1 - y_6$

$15 H_6 + 15 H_2 \rightarrow \frac{15}{6}$

$15 H_5 + 10 H_3 + 5 H_2 \quad \frac{10}{3}$

$15 H_4 + 6 H_4 + 8 H_3 + H_2 \quad \frac{}{3 \quad t}$

x_i

$x_i x_j \quad y_i \rightarrow 6$

$x_i x_j x_k \quad y_i y_j$

$y_1 - y_6 \rightarrow 6 \quad 3 \quad t$

$3 H_6 \quad 3 + 3 H_3$

$t H_{2t} \quad t H_{2t-1} + 1 \quad t H_{2t-2} + 2 \quad t H_t + t$

$k = 2t$

$\binom{2t}{t-1} \quad x_1, \ldots, x_t \quad y_{i_1} - y_{i_{t-1}}$

$\binom{2t}{t-1} \rightarrow y_1 - y_{2t}$

$\binom{2t}{t-1}\left(H_{2t} + H_{t-1}\right) > \binom{2t}{t-1} H_t + \sum \binom{t}{i}\binom{t}{i+1} H_{t-i}$

$\binom{2t}{t-1} H_{2t-1} + \binom{2t-1}{t-1} H_t + \binom{2t-1}{t-2} H_{t-1}$

$\frac{\binom{2t}{t-1}}{2t} > \frac{\binom{2t-1}{t-1}}{t}$

$c \in T$ $\frac{n}{k}$ $|T| \leq |S|$

$\frac{b-c}{(a+b)b} > \frac{1}{(a+b)}$ $\frac{1}{a+1}$

Rammond

$\frac{n}{k}$

$(b-c)$

$> \frac{a+1}{a}$ $\frac{c^2}{c-1}$

$a=1$

$\underbrace{\sum x_i^2}$ $c \in T / W$

c'

$\boxed{\frac{a+c}{a+b}}$ $\frac{a}{k}$ Corex $\frac{a}{k}$

$\frac{n}{k} \geq \frac{k}{n}$ $\frac{k}{n}$

$c' \in W$

c_2 a_1 B $\boxed{E \exists R}$

c' T $\cap$

$\sum a_i = A$ $\frac{a_1+c_1}{A+b}$ $\frac{a_2+c_2}{A+B}$ $\frac{a_n+c_t}{A+B}$ $\frac{B-C}{A+B}$ n/x $k=3$

$\sum c_i = C$

$A+B+C=m$

$A+B=k$

$A+B=k$

$\frac{n}{k}$ $\frac{k}{n}$

T PAV $\frac{1}{}$ $\sum x_i^2$ $[m]/m$

T

$\underline{\sum x_i^2}$

PAV phragmen

c

$\frac{1}{a+b}$ $\frac{\sum x_i^2}{k^2}$

A k

K $\frac{k}{n}$ $\sum x_i^2$

$\frac{k^2}{n^2}$ $1/n$ $1/m$

we have so many candidates

+

5

2^{c} $\binom{m}{c}$

C

$a=2$ ①

1 ✓ mx A_i

$2 \rightarrow$

$T = \{c_1\}$ ① $\frac{b}{a+b}$

$\frac{n}{k}$ $|T| = 1 \rightarrow$ EJR $\frac{1}{b}$ $\frac{a-c}{(a+b)}$ ما کسانی

$|T| = \{c_1, c_2\}$ ①

$\frac{b-c}{b}$

c_1 c_2 c_3

c_2 $|S| \geq \frac{2}{k}$

c_1 c_1 c_2 PAV PAV → phragmen

T_1 T_2

$|A_i| \ll k$ ~~x~~

T $\frac{a}{a+b}$ $\frac{b}{a+b}$

⇒ گروه منسجم و بزرگ افراد

T_1 T_2

T'

EJR

W

$<$ k

گروه بلوکه کننده

T - minimal

کوچکترین گروه بلوکه کننده